

Arivuchezhiyan E

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Executive Summary

First-year M.Tech Integrated CSE student at SSN with hands-on experience building software across systems programming, full-stack web development, and cross-platform mobile engineering. Built and shipped projects spanning a custom OS kernel in C/x86 Assembly, a real-time MERN ride-hailing platform, and an encrypted 12-feature Windows–Android integration suite. Ranked 2nd in XII State Board examinations (571/600) and currently maintaining a 7.656 CGPA.

Education

M.Tech (Int) CSE	Sri Sivasubramaniya Nadar College (SSN)	CGPA: 7.656	2024 – 2029
XII (State)	Paarivallal Higher Secondary School, Ariyalur	Marks: 571/600 (Rank 2)	2023
X (State)	Paarivallal Higher Secondary School, Ariyalur		2021

Projects

Assistant	Python, AI, Automation	2025 🔗
- Built a modular AI assistant in Python integrating speech recognition, task scheduling, and OS-level automation across 5+ independently deployable plugins. - Reduced manual task execution time by approximately 60% through natural language command routing and cross-application automation pipelines.		
OpenContinuity	Electron, TypeScript, Kotlin, AES-256-GCM	2025 🔗
- Designed a 12-feature Windows–Android integration suite (clipboard sync, file transfer, screen mirroring, notifications, battery monitoring) using Electron and Kotlin with sub-50ms local network latency. - Secured all inter-device communication with AES-256-GCM encryption and ECDH key exchange, and automated device discovery via mDNS, eliminating manual IP configuration.		
Uber Clone	MERN Stack, React, Socket.io, Stripe API	2025 🔗
- Built a full-stack MERN ride-hailing platform with real-time GPS tracking (Socket.io + Leaflet.js) and role-based dashboards for 3 user types: customers, drivers, and admins. - Integrated Stripe payment workflows supporting end-to-end transaction processing, ride history, and user rating persistence on MongoDB Atlas.		
Elite OS	C, x86 Assembly, Kernel Development	2024 🔗
- Developed a custom xv6-derived OS kernel in C and x86 Assembly, implementing preemptive process scheduling, virtual memory paging, and 8 custom system calls. - Achieved full usertests.c suite pass rate, validating memory allocator correctness, context-switch integrity, and system call isolation under concurrent load.		

Technical Skills

Languages:	C, C++, Python (Scripting, Automation), Java, JavaScript, Assembly (x86)
Frontend:	React.js, HTML, CSS, JavaScript
Mobile:	Flutter, Kotlin
Backend:	Node.js, Express.js, Firebase
Databases:	MongoDB, SQL
Tools & Platforms:	Git, GitHub, Socket.io, Electron
CS Fundamentals:	Data Structures & Algorithms, OS Internals, Systems Design

Certifications & Courses

Google — Privacy and Security Fundamentals : Certificate	NPTEL — Affective Computing (In Progress)
Embedded Development Workshop — Hardware Systems : Certificate	

Clubs and Activities

SSN Coding Club (UI/UX Domain Member) IEEE Computer Society SSN (Volunteer)	2025 – Present
NSO Volunteer (Inter-college Sports Coordinator) Hackathons (Participated at SNU & St. Joseph's)	